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Analyzing Harms from AI-Generated Images and Safeguarding Online Authenticity

The democratization of image-generating artificial intelligence (AI) tools without regulatory guardrails has amplified preexisting harms on the internet. The emergence of AI images on the internet began with generative adversarial networks (GANs), which are neural networks¹ containing (1) a generator algorithm that creates an image and (2) a discriminator algorithm to assess the image's quality and/or accuracy. Through several collaborative rounds between the generator and discriminator, a final AI image is generated (Alqahtani, Kavakli-Thorne, and Kumar, 2021). ThisPersonDoesNotExist.com, a site created by an Uber engineer that generates GAN images of realistic people, launched in February 2019 to awe-struck audiences (Paez, 2019), with serious implications for exploitation in such areas of abuse as widespread scams and social engineering. This was just the beginning for AI-generated images and their exploitation on the internet. Over time, AI image generation advanced away from GANs and toward diffusion models, which produce higher-quality images and more image variety than GANs. Diffusion models work by adding Gaussian noise² to original training data images



through a forward diffusion process and then, through a reverse process, slowly removing the noise and resynthesizing the image to reveal a new, clean generated image (Ho, Jain, and Abbeel, 2020). Diffusion models are paired with neural network techniques to map text-to-image capabilities, known as *text-image encoders* (Contrastive Language-Image Pre-training [CLIP] was a milestone in this space), to allow models to process visual concepts (Kim, Kwon, and Chul Ye, 2022). Thus, the commercialization of diffusion models (DALL-E, Stable Diffusion, Midjourney, Imagen, and others) put the power of synthetic image generation in the hands of the user on a global scale. The rise of image generation tools has introduced synthetic forms of such safety harms as mis- and disinformation, extremism, and nonconsensual intimate imagery (NCII), causing further disarray and damage in the internet ecosystem.

Abbreviations

AI	artificial intelligence
C2PA	Coalition for Content Provenance and Authenticity
CAI	Content Authenticity Initiative
CID	civil investigative demand
CLIP	Contrastive Language-Image Pre-training
CSAM	child sexual abuse material
FTC	Federal Trade Commission
GAN	generative adversarial network
GIFCT	Global Internet Forum to Counter Terrorism
NCII	nonconsensual intimate imagery
NIST	National Institute of Standards and Technology
OECD	Organisation for Economic Co-operation and Development

The societal harms from AI image generation tools have yet to be effectively addressed from a regulatory standpoint because of a nexus of policy challenges, such as copyright protection, data privacy, ethics, and contractual requirements. Recent fears rising from generative AI have somewhat moved the policy needle on AI regulation, sparking great interest in Congress and the executive branch (as evidenced by the Senate AI insight forums and President Joe Biden’s executive order on AI, respectively [White House, 2023b]). However, without legislation addressing safety and societal issues related to generative AI, executing a coherent regulatory strategy to address the harmful effects of AI-generated images on the internet is a tall order. (As of this writing, Section 230 of the Communications Decency Act of 1996 serves as the sole piece of legislation for internet regulation [U.S. Code, Title 47, Section 230].)

In this paper, I delve into safety harms and challenges from AI-generated images and how such images affect authenticity on the internet. The first section outlines the role of image authenticity on the internet. In the second section, I review the technical safety challenges and harms for the image generation space, then look at industry solutions to authenticity, including the promise of provenance solutions and issues with implementing them. The third section outlines several policy considerations to tackle this new paradigm that largely focus on provenance, given its promise as an authenticity solution and relevance in policy conversations. In this paper, *content authenticity* in the context of images refers to establishing transparent information about images (both human and AI-generated), whether in origin, context, authorship, or other areas, in a way that is accessible to users on the internet. Throughout, this paper focuses

on content authenticity, as it could play a key role in shaping public trust in image content broadly and covers a wide swath of issues, from disinformation to synthetic NCII.

The Role of Image Authenticity on the Internet

Images have played an important role in the history of the internet, informing people about current events, sparking emotional responses to war atrocities and injustice, galvanizing individuals to support a cause, and much more. Research shows that people tend to respond more viscerally to images than they do to text online (Medill School of Journalism, 2019). Studies suggest that the brain processes visual stimuli more rapidly than it does words (Alpuim and Ehrenberg, 2023). Furthermore, images can increase a viewer's perception of the truthfulness of an accompanying statement (Alpuim and Ehrenberg, 2023). However, the era of treating images as “proof” from a social standpoint is rapidly changing.

Image authenticity on the internet is in jeopardy, as AI-generated images without *proof of provenance*, or the origin of a given image, are affecting how people perceive current events and public figures. The issue is not only a decrease in content authenticity but also a lack of knowledge and tools among many users to help navigate this paradigm shift in the information domain. Opportunistic actors are taking advantage of accessible AI tools to reduce trust in content and media, particularly during tumultuous periods, such as violent conflicts. An example of this is the inflammatory AI-generated images spawned after Hamas's October 7, 2023, attack in Israel and the subsequent conflict in Gaza. Multiple AI-generated images

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(some photorealistic) spread widely on the internet, conjuring fake crowds of Israelis marching in support of the Israeli government or unusual images of Gazan children in the midst of explosions; these images were rarely labeled as AI-generated (Klee and Ramirez, 2023).

Solutions to this problem that focus on content authenticity could be the most valuable by providing individuals with more transparency about content that they consume and, therefore, more agency in terms of how to interpret that content. For example, solutions that provide metadata information to the user, such as authorship, geolocation data, and tools used in the editing of an image, can all be useful for user interpretation. Though the research behind authenticity measures affecting user trust in content is not conclusive, and content authenticity is not a silver bullet for solving all AI-driven harms on the internet, content authenticity solutions are a step in the right direction.

Image-generation technology will continue to evolve, become more advanced, and likely become more photoreal, increasing the escalation potential for harm. There is no foolproof way to make these models entirely safe for use. To start solving issues of AI-generated images fueling disinformation, harmful propaganda, NCII, and other safety harms, the United States must first develop solutions in regard to authenticity and improve the public's access to information about these images.

The first step in navigating this shift is to comprehend how AI image generation tools produce safety issues and challenges and how current safeguards are insufficient to tackle the problem.

Safety Challenges in Image Generation

Safety challenges in the AI image generation landscape begin at the technical level, and the most significant safety

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challenges are due to biases and harms from training data, the existence of open-source image models, and a piece-meal approach to content moderation at the user level. The current AI image generation space mainly consists of text-to-image diffusion models, such as Midjourney, DALL-E, and Stable Diffusion, that generate images based on user prompts. Understanding the fundamentals of safety issues in text-to-image diffusion models can show why these models can produce unsafe images. Furthermore, diving deep into training data, open-source models, and content moderation reveals that these technical mitigations are simply not enough to prevent the generation of harmful content, and the United States needs authenticity solutions to manage risks and harms.

Image generation models reflect societal and representational biases on the internet because they are trained on data scraped from the internet. For example, there are far more images sexualizing women on the internet compared with similar images of men and far more images of men in professional roles (doctor, lawyer, engineer, etc.) than similar images of women. Image models conceptualize these representational biases and have become quite adept at generating content that both oversexualizes women and highlights men in professional positions (Heikkilä, 2022). Recent research shows that there is still much work to be done to make image models safer and less biased, as there are still severe occupational biases in models, which result in the exclusion of groups of people from generated results (Naik and Nushi, 2023). Furthermore, *Rest of World* in 2023 conducted an experiment with Midjourney that showed that the tool typically represented nationalities using harmful stereotypes: Images of a “Mexican person” mainly showed a man in a sombrero, and images of New

Delhi almost exclusively showed polluted streets (Turk, 2023). The internet is inherently biased, given the nature of its human inputs: When you scrape highly biased data, you will generate it as well.

Safety harms in these models also stem from training data. To start, data labeling is largely outsourced to providers that specialize in scaled labeling, which is cost-effective. However, this process can introduce biases and inaccuracies in human labels for training data (Smith and Rustagi, 2020). When developers scrape data from the internet for image generation, their main method to ensure that models are safe is to filter training data and attempt to reduce the prevalence of harmful content in the training process (Smith and Rustagi, 2020). However, this method is contingent on how effective these filters are in rooting out harmful content while ensuring that they are not too conservative in excluding training data so that the models are still trained on a wide range of data and retain quality and creativity in their generations. Furthermore, even with robust safety filtering, the ability for models to deduce concepts across different types of benign images can result in the generation of a harmful image. For example, an image model that is trained on images of beaches but not on pornography will understand how the human body looks with swimsuits or minimal clothing. The same model could also be trained on images of children going to school, playing outside, etc.—all of which are benign images. These model capabilities combined (without further safety measures) will likely allow the model to create images of scantily clad or even nude children through malign prompt engineering techniques. However, most developers would still want training data to have images of beaches and children to ensure high quality and effective generations. Unfortu-

nately, the trade-off between safety and what constitutes a quality product is often difficult.

The discussion of safety challenges with image generation would be incomplete without highlighting the open-source space for this critical technology. The debates around open-source generative AI are abundant: Those who are in favor highlight the benefits of open access in improving models and their safety capabilities with wider researcher access, and those against are focused on the potential for malign use and challenges with monitoring and controlling model use. With image generation in particular, open access has benefited malicious actors in removing safeguards and generating harmful content at scale with the use of fine-tuned open-source models. For example, when Stable Diffusion was open-sourced in 2022, Unstable Diffusion was born. Unstable Diffusion was a large server that used Stable Diffusion’s open-source model with reduced safeguards to create not-suitable-for-workplace content (Wiggers and Silberling, 2022). An even more alarming example is CivitAI, a site created for AI image generation that allows users to browse thousands of models in order to generate pornographic content and synthetic child sexual abuse imagery, streamlining the nonconsensual AI porn economy (Maiberg, 2023a). A more recent example is a Stanford Internet Observatory investigation that found hundreds of images of confirmed child sexual abuse material (CSAM) in an open dataset called LAION-5B, which is commonly used in image generation models, such as Stable Diffusion (Thiel, 2023). The use of open-source image generation poses ethical concerns related to child exploitation, consent, and fair use. From a consent and fair use perspective, such use could disproportionately affect sex workers, who may have their images

scraped from the internet during the model training process, allowing their likeness to be reproduced synthetically without their consent and without compensation. Despite the benefits of open-source software ensuring greater access to image models, it has accelerated safety harms, especially those related to sexual content and consent.

Last, it is valuable to note the content moderation safeguards of classifiers and keyword-blocking, which are used in many image generation tools, and to detail why they are insufficient for safety. Safety, as a first principle across the AI development life cycle, is the path to ensuring safe image generation, starting from filtering data prior to training. When the majority of safety work is done in a monitoring capacity through content moderation, harmful generations are likely to fall through the cracks. The current image generation models on the market cannot entirely prevent the generation of NCII content, synthetic CSAM, extremist propaganda, and misinformation. Some of them, such as Stable Diffusion, rely on a weak form of content moderation: blocking specific keywords from being used. Stable Diffusion enacted keyword-blocking for words related to the human reproductive system to attempt to prevent the generation of pornography (Heikkilä, 2023). Bing’s AI Image Generator, powered by DALL-E 3, blocked the keywords “twin towers” to prevent any harmful depictions of 9/11 (David, 2023). Despite keyword-driven safeguards, bot accounts were able to spread an AI-generated image of the Pentagon on fire, which briefly went viral (Bond, 2023). Keyword-driven moderation is at best a piecemeal solution that only helps to moderate low-hanging fruit and can hardly cover the multitude of harmful narratives at scale. Another form of moderation can be done through classifiers, such as *prompt input clas-*

sifiers that identify violative prompts and *image output classifiers* that classify images that should be blocked, both of which are used in DALL-E 3 to prevent harmful generations (David, 2023). Although such moderation is more holistic than just keyword-blocking, these classifiers are not foolproof. Specifically, red-teamers for DALL-E 3 found that (1) the model refusals on disinformation could be bypassed by asking for style changes, (2) the model could produce realistic images of fictitious events, and (3) public figures could be generated by the model through prompt engineering and circumvention (OpenAI, 2023). Broadly, content moderation, through both classifiers and keyword-blocking, was never built to infer context, and an attempt to do so is practically an impossible task. For example, users can attempt to bypass many keyword and classifier safeguards by using visual synonyms, such as “flesh-colored eggplant” (for a phallic image), “red liquid” for blood, and “skin-colored sheer top” to generate nudity; the list becomes endless.

These harms and safety challenges are shaping content authenticity on the internet. Though much of the spotlight of content authenticity is on mis- and disinformation, the existence of synthetic NCII, synthetic extremist propaganda, and more also directly shapes users’ perceptions of reality and can cause great harm to individuals and societies. Unfortunately, from a technical perspective, eradicating all these safety harms seems unlikely, given the challenges of content moderation, bias and safety issues with training data, and the existence of tailor-made open-source models. Instead, the United States should look to create greater transparency around these images through solutions that promote access to information about the origin of content. These solutions may be less fruitful to mitigate

synthetic NCII content, given that the core of mitigating NCII is not just deciphering whether the content is AI-generated but also ensuring legal recourse and accountability for individuals and entities that distribute and create such content. However, authenticity solutions such as provenance and watermarking could help dissuade and disincentivize malicious actors from using tools that adopt these safeguards and help debunk photorealistic NCII, disinformation, and much more.

Framing this issue as one of content authenticity could empower users and put the onus of responsibility in terms of securing transparency and accountability for safety issues in image generation models on technology providers and government entities that can enforce regulations to combat these harms.

Industry Solutions to Preserve Image Authenticity

The issue of preserving authenticity on the internet is not new. Fake accounts, bots, phishing emails, and more have been persistent issues for years. Challenges with deepfakes started to spark more-deliberate conversations about photo and video authenticity; in 2017, a Reddit user exploited Google's open-source deep-learning library to post pornographic face-swap images (Adee, 2020). Now, the fight to preserve authenticity has become even more crucial, given a lack of policy safeguards and sufficient platform-level enforcement, as well as the speed at which AI image generation is improving. When examining the issue of harms caused by AI-generated images through the lens of "authenticity," it is important to think through what kinds of tools and technologies will best support individuals in

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determining whether an image is authentic, despite adversarial motives. A user-centric approach to this issue is key, given that the success of content authenticity initiatives can be shaped by user experiences, personal bias, and/or beliefs about technology.

Watermarking, Hashing, and Detection

There have been several debates about the right approach to authenticity. The concept of *watermarking* has dominated the conversation and is a term that the White House included in its voluntary AI commitments for industry (White House, 2023a). However, there is confusion and a lack of consensus in the field about the definition of a watermark and how effective watermarks can be. A helpful way to define watermarking is provided by the Partnership on AI: a form of disclosure that can be visible or invisible to the user and includes "modifications into a piece of content that can help support interpretations of how the

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content was generated and/or edited” (Partnership on AI, 2023). Watermarking (both visible and invisible/metadata-based) can be a useful disclosure for the general public for content interpretation, but it is far from a holistic solution, given the need for watermarks to be robust against adversarial attacks, to overcome challenges to secure widespread adoption, and to be understandable by a consumer or user across different social platforms and hardware devices.

Another approach in the field is *hashing*, or fingerprinting image content, which happens after an image is created. *Cryptographic hashing* is used to determine exact matches, whereas *perceptual hashing* can find similar matches that may not be exactly the same image (Ofcom, 2022). The merits of hashing have been particularly evident in the identification of CSAM and terrorist content. For example, the National Center for Missing and Exploited Children and the Global Internet Forum to Counter Terrorism (GIFCT) are nonprofit organizations that use hash-sharing platforms with technology companies of thor-

oughly vetted hashes of confirmed CSAM (National Center for Missing and Exploited Children, undated) and terrorist content (GIFCT, undated), respectively. Hashing can prove useful in terms of sharing content among social media platforms that very clearly belongs in specific categories of abuse, but it is less practical for use across a variety of content, given that hashing occurs retroactively and cannot be done well at scale. It is also susceptible to adversarial attacks and is vulnerable to database integrity issues and discrepancies caused by human review in the content attribution process (Ofcom, 2022).

Last, an approach that has been discussed for several years is *detection*. Both established companies and smaller startups—such as Intel (Clayton, 2023), Optic (Kovtun, 2023), and Reality Defender (Wiggers, 2023)—have produced deepfake detection solutions. Though the technology can be promising, it comes with a host of issues. Traditionally, in the cybersecurity space, detection and evasion are a cat-and-mouse game, with detection needing to constantly improve as both adversarial actors and the technology itself improve. Reality Defender’s chief executive officer claims that provenance and watermarking solutions are weaker, given that they require buy-in, and that Reality Defender’s product, which is focused on inference (determining the probability of something being fake), is a more robust solution (Goode, 2023). However, even with high rates of efficacy, the onus would still be on users to gauge how much they should trust a piece of content based on a probability metric alone. Furthermore, current image detection capabilities have accuracy issues, as reported in a *Bellingcat* investigation. *Bellingcat* assessed a tool by Optic (called “AI or Not”) and determined that it was successful in identifying AI-generated and real images quite accurately, except

when AI-generated images were both compressed and photorealistic, in which case accuracy dropped significantly (Kovtun, 2023). Image compression (a relatively common practice) can assist malign actors in evading detection, especially on social media platforms, which generally compress all uploaded images. Detection tools are not foolproof and are fragile to minor perturbations.

Provenance

Though no individual solution to content authenticity is holistic, *provenance* is emerging as a useful tool to proactively preserve origin metadata and/or any editing or changes to a given piece of content. Detection methods risk being less useful as technology continues to improve and evolve while placing the onus of using detection tools on a user every time they come across content that they deem to be suspicious. Furthermore, such methods can lack accuracy, further obfuscating the decisionmaking process for an individual to assess content for its authenticity. Provenance approaches, such as establishing the origin of a piece of content through secured metadata, are generally more robust, given that they focus on the origin of content rather than proving whether something is real or fake. Furthermore, when implemented well, provenance solutions can be incorporated across the content supply chain—in AI image generation tools, social media platforms, news sites, and more—so that metadata information is readily available to a user and can complement and be used in tandem with watermarking and fingerprinting initiatives.

Examining the original approaches to provenance starts with understanding the history of data creation and modification, both of which are useful for AI image gen-

eration (Zhang, Chapman, and LeFevre, 2009). In an AI image generation context, provenance could map the origin of an image through a cryptographic hash or signature that is applied and attached to the content, is stored securely through encryption, and is “tamper-evident” or is able to show whether the image has been altered in any way. Metadata information, available to users through labels, could also help define the trustworthiness of an image. In practice, making provenance a success goes far beyond a cryptographic signature and requires the widespread adoption of one or many interoperable frameworks across different mediums: hardware (e.g., cameras, smartphones), editing software (e.g., photo editing programs, face-swap apps), and publishing and sharing entities (e.g., news media, social media platforms), which is a very challenging task in practice.

The industry ecosystem has rallied around the application of provenance to support content authenticity ecosystems for AI-generated images. Industry leaders include Adobe, Intel, Microsoft, and Truepic, all of which are members of the Content Authenticity Initiative (CAI) and the Coalition for Content Provenance and Authenticity (C2PA). Both groups are focused on cross-industry participation to tackle the issues of media transparency and content provenance, with the C2PA framework underlying many of these initiatives, and have released products that use the C2PA framework. For example, Adobe launched its “Content Credentials” feature in 2023, which uses the C2PA standard to allow the attachment of secure, tamper-evident metadata on an export or download (Quach, 2023). The C2PA framework is an interoperable specification that “enables the authors of provenance data to securely bind statements of provenance data to instances of content using

their unique credentials” (C2PA, undated-a). C2PA also provides a “chain of provenance,” by which it can track whether an asset was created from a bevy of different assets. For example, the author or creator of a video (which may include AI-generated images, deepfake audio, and other varied forms of media under the C2PA specification) can tag the provenance of these various assets within the video and can determine what is and is not included. The creator of the content can also redact information to protect their privacy and/or security. C2PA also functions as an opt-in, open framework, meaning (when applied in good faith) it requires the permission of its users (likely for privacy reasons) and is built for any entity to adopt it (Castellanos and Gregory, 2021). It is valuable to note that C2PA has gaps in which privacy could be breached by malicious actors, as claim generators could require users to add sensitive information to manifests³ for a given image (C2PA, undated-b).

Outside C2PA, Google has taken its own approach to securing authenticity online through its “about this image” feature, which provides background information on a given image, including when the image was first indexed by Google, where it appears online, and where it may have first appeared (Goode, 2023). In practice, this feature is helpful for users who wish to source content, but it is not a clear and reliable provenance method, given that it does not collect tamper evidence or provide thorough provenance metrics. Instead, this approach can be seen as complementary to C2PA efforts and is one method to empower user literacy for AI-generated images.

Issues for Provenance

Despite the developments in provenance, challenges remain: adoption across the content supply chain (given

resource concerns for some entities), challenges with implementation and privacy for social media platforms, and broader privacy concerns across the globe. C2PA is much less useful when the necessary organizations across the supply chain for image creation and dissemination do not adopt it. For example, if an original image was developed in an image generation tool that adopted the C2PA specification and was subsequently changed in an editing tool that did not use C2PA, the provenance metadata of the image would only show the origin data for the original image, not its subsequent editing. Some entities across the supply chain of content (smaller media companies in the Global South, for example) may not have the resources to adopt a provenance specification and integrate it throughout their products. Furthermore, without the adoption of an interoperable specification by social media companies, users will not be able to get consistent provenance information on news-related images, political images, and other content across the web. Implementation is a challenge for social media platforms; currently, most platforms strip image metadata when a user posts an image on a site (Laurent, 2013). This practice is largely for privacy reasons. Digital image metadata or EXIF data can reveal information about the person who took an image, such as a given image’s geolocation (Marin, 2020). This paradigm of stripping metadata would have to change for provenance data to be available to users, and this change must be done in a way that allows users to opt in. Last, there are considerations to be made about how privacy could be infringed by the weaponization of provenance metadata, and frameworks such as C2PA could be abused to enforce journalistic identity and enforce laws restricting freedom of expression, particularly in repressive states (Castellanos and Gregory,

2021). Broader privacy concerns could also cause hesitancy for provenance adoption globally, especially for media entities in nondemocratic states.

The ideal user experience for image authenticity would involve provenance information being available for image content from a user's phone or camera, news sites, photo editing tools, social media platforms, and other publishers in a manner that uses opt-in to protect user privacy and is interoperable. Realizing this ideal is likely to be a Herculean undertaking. However, even if multiple providers on the internet do not opt in to the same interoperable specification, AI image generators and social media platforms at the very least must work together and use a common provenance standard (such as C2PA) to enable user agency within social spaces and provide transparency to users about metadata tagged in AI-generated images. Public trust in image content can only be improved with greater accessibility of provenance information for users to decide how to perceive a given image.

Policy Considerations

The 2023 White House–led voluntary AI commitments include the need for the development of “robust provenance” and “watermarking” efforts by AI companies that have opted in to the commitments (White House, 2023a). Voluntary commitments and self-regulation are not enough to hold private entities accountable; however, striking the balance between regulation and voluntary efforts is a notable challenge. The voluntary nature of these commitments has some benefits, such as greater private-sector cooperation, information-sharing, and room to evolve

transparency mechanisms and policies on provenance and watermarking. However, without common and universal standards on provenance and/or watermarking for companies to abide by and without forms of social media regulation such as transparency reporting, among other methods, policy solutions for image authenticity will be a toothless endeavor. Though the harms of AI-generated content cannot be erased, they can be mitigated through interoperable information disclosures on images. The most effective policy solutions will focus on building public understanding related to AI-generated images and transparency about content authenticity through collaboration between government and industry, as well as government-led accountability mechanisms to hold technology companies accountable. Several policy steps can help tackle the issue of harms stemming from AI-generated images, using an authenticity-focused lens.

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The National Institute of Standards and Technology Must Lead U.S. Standards on Provenance and Watermarking

The United States needs concrete and enforceable standards on provenance and watermarking for industry adoption. Beyond the National Institute of Standards and Technology (NIST) AI Risk Management Framework (RMF), the United States does not have concrete government standards on AI. NIST's AI RMF is a promising start on a long road in the development of robust and adaptable AI standards but is too vague to be directly implemented on specific generative AI issues, such as harms from AI-generated images. NIST helps develop standards through consensus-building and working with organizations that create standards but does not issue standards on its own. However, it can be a key driver for synthetic content standards in the United States and on the international stage. NIST is working on generative AI profiles via its Genera-

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tive AI Public Working Group and NIST AI consortium and must build specific standards and best practices on provenance and watermarking to create a universal U.S. standard to be implemented in regulation (NIST, undated). Furthermore, President Biden's 2023 executive order on AI designates NIST as the United States' epicenter for AI safety and security standards, under which NIST is designated to develop clear standards for digital authenticity and synthetic content for U.S. government adoption efforts (White House, 2023b).

Designing standards around provenance and watermarking adoption is not a simple task; the design must provide flexibility and room for evolution, given that these standards will not be immediately enforced by a regulatory body. Standards should be centered on NIST's trustworthiness principles (NIST, undated) and measurement science and should focus on measures that empower the public's consumption of images, both human and AI-generated. Measurement science will be crucial to evaluate the efficacy of technical solutions, such as forensic techniques, interoperable provenance frameworks, digital identity verification, and image forgery detection. Sociotechnical valuations guided by measurement science could also help the U.S. government understand how individuals consume content that is AI-generated and human-generated in different settings and how provenance affects the perception of both kinds of content. Standards should be guided by robust measurement science, and vice versa, to evolve and improve in a symbiotic manner. For example, a standard could designate what is deemed to be a robust watermark: a digital signature that is tamper-evident, that cannot be easily cropped out or removed, and that is securely applied

to an image. Standards on provenance could also specify that tools with provenance features must be opt-in and transparent to the user to protect privacy and data rights. There likely should also be a documentation standard, in terms of how provenance data are documented and secured, to prevent adversarial attacks. These are just a few examples of what standards, shaped by measurement science and evaluations, could look like at NIST. To determine the efficacy of standards, NIST could use experimentation and testing methods for these standards in a regulatory sandbox, in which firms could test the implementation of potential standards internally in collaboration with NIST, and NIST could determine the efficacy of standards based on results (Organisation for Economic Co-operation and Development [OECD], 2023). Regulatory sandbox exercises in the financial technology industry, as reported by OECD, have helped garner an evidence-based approach to regulation and policy (OECD, 2023). After standards are finalized, the U.S. government should deploy these standards both in technology regulation via Congress (especially for large AI developers that have already made voluntary commitments on AI) and for internal government use. The latter will likely be less challenging to implement from a political perspective, given intergovernmental incentives for adoption as a trust mechanism for U.S. government content and the absence of a profit motive to interfere with the adoption of standards. The development of these standards will not exist in a vacuum, and NIST is well-positioned to work with civil society, industry, and other federal agencies to shape the future of provenance and watermarking and help the public navigate authenticity on the internet.

Tripartite Engagement on AI-Generated Image Harms Is Crucial

Currently, industry actors are leading the conversation on AI-generated image harms by releasing and pioneering detection tools, forming coalitions (such as CAI and C2PA), and adopting provenance and/or watermarking solutions. The proactiveness of industry on this issue, likely affected by the White House's voluntary AI commitments, is promising, but there are no formal engagement bodies or working groups among industry, the U.S. government, and civil society organizations on AI-generated image harms and authenticity issues. A working group that includes relevant U.S. government, civil society, and industry stakeholders will go a long way toward anticipating emerging risks in the AI image space, helping the federal government create and implement enforceable standards, and ensuring information-sharing on these issues. The United States is on the verge of the 2024 presidential elections, thus elevating the need to tackle authenticity issues from AI-generated images and ensure that the public is informed about the content it is consuming, which is where industry and civil society come in. These entities should not operate in silos. The Department of Commerce, in coordination with NIST, must formally engage industry and civil society actors on these issues through a working group to increase transparency about authenticity solutions (provenance, detection, etc.) that are being explored at technology companies, with the three branches of government incorporating these insights to inform future standards and practices that would best benefit the public. The right pathway to designing effective regulation involves staying ahead of the technology curve, which will only be possible if there is

open engagement with industry on these issues through a joint working group with representatives from government, civil society, and industry.

The Federal Trade Commission Should Continue Consumer Protection Actions for AI

The Federal Trade Commission (FTC) must get involved with issues related to AI-generated images and how they affect market competition and consumers on the internet. The FTC has taken an active interest in AI, highlighting consumer concerns and opening an active investigation on OpenAI over data leaks and inaccuracies in ChatGPT (Zakrzewski, 2023). Furthermore, the agency issued an omnibus authority for “compulsory process” for AI products and services in nonpublic investigations, which allows the FTC to issue legally binding civil investigative demands (CIDs) to “obtain documents, information and testimony that advance FTC consumer protection and competition investigations” for the next ten years (FTC, 2023). This development could be highly advantageous in terms of gaining more information from large AI developers and other AI companies about harms created or exacerbated by their systems.

To build on these efforts, the FTC must focus its investigation on harmful image generation tools, such as those that create nonconsensual intimate AI images, and impose costs on businesses that are directly profiting from consumer harm, such as any companies that are profiting from nonconsensual AI pornography (Maiberg, 2023b). By imposing financial costs and issuing CIDs on harmful open-source image generation tools, for example, the FTC can help reduce authenticity-driven harms and

dissuade private actors from profiting off of these tools. Furthermore, the FTC should generally use its CID authority to investigate harms from AI image generators and collect detailed information and documentation of consumer harms from these companies to better understand the macro state of safety harms from image generation and potential harms to market competition. The agency should also conduct an internal research study on consumer harms that can occur as a result of deceptive activities involving AI-generated images. The results should be shared externally with other federal agencies and the Executive Office of the President to better shape next steps to understand future market and consumer risks that could result from a lack of image transparency on the internet (FTC, 2023), including, for example, the adverse effect on public trust when AI-generated images with no authenticity disclosures are used in deceptive advertising to shape consumer opinion. Though the FTC often gets politicized in Congress, the agency is well-positioned to tackle harms from deceptive activity and to evaluate potential harms to consumer protection with respect to AI-generated images.

The United States Is Overdue for Transparency Regulation for Social Media

The U.S. government needs to develop regulations on AI-generated content on social platforms to safeguard users and society from the harms exacerbated by AI-generated images. These regulations would take the form of transparency requirements and accountability mechanisms for regulatory compliance.

Social media regulation in the United States is ostensibly inadequate, given a sheer lack of regulation beyond

Section 230 of the Communications Decency Act, the effectiveness of which has been debated by members of Congress across the political spectrum. However, any future regulation of AI will be incomplete without the regulation of social media platforms on AI-generated content. Part of the challenge with enforcing new legislation and regulations is the boundaries created by Section 230, which dictates that platforms cannot be treated as publishers or broadcasters and therefore cannot be held legally accountable for the content that proliferates on their platforms (Matthews, Williams, and Evans, 2023). Though the debate on repealing Section 230 is long and complicated, there are actions that Congress can take within the current legal status quo.

First, transparency requirements are essential. The lack of transparency from social media platforms has led to much legislation in Congress but little that has been passed to enact regulation. The 2022 European Digital Services Act introduced new forms of compliance for platform transparency disclosures and has already resulted in increased openness and transparency. For example, TikTok opened its application programming interface (API) for access to European academics and researchers and provided options to disclose commercial content through labeling, among other initiatives (Lomas, 2023). The United States should follow Europe's example on transparency legislation and specifically mandate that very large online platforms (also known as VLOPs) publish biannual (twice per year) transparency reports on the amount of harmful AI-generated images disseminated, how much of that content was taken down, trends related to those particular harms (disinformation, sexual content, etc.), how many users engaged with confirmed AI-generated mis-

The lack of transparency from social media platforms has led to much legislation in Congress but little that has been passed to enact regulation.

or disinformation, how much AI-generated content was political disinformation, and in which geographic locations these harms originated. Though large platforms have their own versions of transparency reporting on a yearly basis, there is no uniform standard for what is reported, much less guidance on what kinds of disclosures should exist. The aforementioned categories for transparency disclosure would (1) represent a meaningful first step toward better understanding AI-related harms on social media platforms, (2) inform the public of the breadth of the problem, and (3) encourage platforms to build robust trust and safety teams to tackle these issues.

Second, the U.S. government should pass regulation that would require social media platforms (in collaboration with large AI developers) to use provenance and/or watermarking disclosures that are easily accessible to users and, ideally, interoperable. The purpose of this legislation and regulation should not be to determine exactly how provenance would be implemented in products; technol-

Making provenance interoperable will require widespread adoption by social media platforms.

ogy companies are best positioned to determine these details. Rather, it should require provenance disclosures for AI-generated images based on future standards created by NIST and guarantee that users have the ability to view provenance data on AI-generated images, especially those that are deemed to contain political content or other sensitive imagery (such as sexual content). As previously mentioned, making provenance interoperable will require widespread adoption by social media platforms, AI developers, and other entities. Some entities will not opt in to provenance frameworks because of resource-related reasons or privacy concerns or because tools are being used for malicious purposes. One solution could be open-source tools that do not include provenance metadata. Though the open-source challenge can be daunting, disclosures on trusted content can help individuals and the broader public retain agency and the ability to interpret what content to trust. The success of this regulation will be contingent on the standards developed and whether they contextualize to different use cases, such as a social media platform versus an AI developer.

Third, existing social media regulation—specifically, the Platform Accountability and Transparency Act (PATA)

and the Digital Services Oversight and Safety Act—must be pushed forward and implemented to get ahead on transparency and accountability actions for technology platforms. PATA will be vital for transparency efforts, given that it requires large technology platforms to provide external platform access to vetted researchers, who submit research proposals approved by the National Science Foundation (Perrino, 2023). Furthermore, PATA imposes costs: If platforms fail to comply, they face actions from the FTC and the potential loss of Section 230 protections, which would come with high costs (Perrino, 2023). This policy solution would increase the public’s access to knowledge regarding generative AI harms, such as disinformation harms from AI-generated images. The Digital Services and Oversight Act mandates that large social media platforms execute risk assessments and mitigation audits to prevent systemic risks, such as content that has a negative effect on “civic discourse, electoral processes, public security,” and many other categories (U.S. House of Representatives, 2022). This large mandate is written to include generative AI harms, given that AI-generated images can be used to manipulate civic discourse, for example. These two pieces of legislation provide the initial foundational action to help curtail risks from generative AI on social platforms and increase transparency to the public of the harms on these platforms. Though there is not likely to be effective Section 230 reform soon, progress on the issue of AI-generated images and their effect on authenticity on the internet is still possible. These actions are not comprehensive but represent progress in the journey toward garnering greater public trust related to the authenticity of AI-generated images.

Conclusion

Developing policy solutions for AI-generated images on the internet may appear to be a narrow endeavor in the grand scheme of all generative AI harms and risks to society. However, examining this niche issue can serve as a useful case study about the future of AI and how it can be shaped to empower people. The harms from AI-generated images—NCII, disinformation used during conflicts or to ruin reputations, harmful political propaganda created to galvanize fringe communities—are ongoing. A fundamental issue that covers all these harms is authenticity and the public’s access to trustworthy information about AI-generated images. The benefits of generative AI will be concentrated and its harms will be diffused if people are left without the right tools to navigate the unencumbered growth of this technology. Industry and policy efforts that center on public trust and authenticity, such as the standards and implementation of provenance methods, will prove to be the most sustainable and most advantageous for society in the long run.

Notes

¹ A *neural network* is a subset of machine learning that uses algorithms to identify relationships within a dataset in order to mimic how the human brain works.

² *Gaussian noise* is a type of statistical noise that is characterized by its probability distribution.

³ C2PA defines *manifest* as “the set of information about the *provenance* of an *asset* based on the combination of one or more *assertions* (including *content bindings*), a single *claim*, and a *claim signature*” (C2PA, undated-c; emphasis in original).

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About This Paper

This paper addresses the myriad safety harms that arise from artificial intelligence (AI)-generated images and what these images mean for authenticity on the internet writ large. The issue of authenticity on the internet is not a novel problem; however, the issue has become exacerbated by generative AI models, such as image generation tools. The goal of this paper is to highlight the challenges of tackling this pervasive harm and to provide considerations for U.S. policy to safeguard the authenticity of images on the internet in terms of this specific issue. The U.S. government has several potential policy levers to address this problem, and the most salient center on securing authenticity on the internet and, by extension, restoring public trust online. This paper should be of interest to policymakers, AI platform developers, and the public.

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